

Factor Investing in Sovereign Bond Markets: Deep Sample Evidence

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Abstract

We examine government bond factor premiums in a deep global sample from 1800 to 2020 spanning the major markets and maturities. Bond factors (Value, Momentum, Low-risk) offer attractive premiums that do not decay across samples, are persistent over time, and consistent across various market and macroeconomic scenarios. The factor premiums diversify to each other, as well as to bond or equity market risks. A combined multi-factor bond strategy provides the strongest risk-adjusted returns. These results strongly show a consistent added value of government bond factor premiums over a passive bond portfolio.

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JEL codes: G11, G12.

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1. Introduction

Several studies show factor premiums are persistent phenomena in markets. Many of these studies examine equity factors (e.g., Fama and French, 1992, 2015, Blitz, 2012), however recently several papers also show individual factors to work well in credit markets (e.g., Houweling and Van Zundert, 2017) or across asset classes (e.g., Baltussen, Swinkels and Van Vliet, 2021). Meanwhile, the size of factor investments has grown tremendously in the industry.¹ However, to date relatively less is known about factor premiums in government bonds markets with investors having been slower to adopt factor investing. This while government bonds are one of the major asset classes in the world, with their size representing about 30% of overall market capitalizations across asset classes (Doeswijk, Lam, and Swinkels 2020). Which factors are present in government bond markets? Are they persistent over time? What is their added value? In this paper we extensively examine global bond factor premiums and answer these questions over a deep sample spanning 221 years across the major government bond markets.

To date, relatively few studies examine government bond factor premiums. Asness, Moskowitz, and Pedersen (2013) were among the first to investigate the use of value and cross-sectional momentum in asset classes other than equities, including government bonds. Kojen, Moskowitz, Pedersen, and Vrugt (2018) demonstrate that carry explains the cross-section of returns in numerous asset classes, including government bonds. These studies examine government bond factors typically over a limited sample, focus on one particular factor, or only consider a long-short perspective. For example, the aforementioned papers roughly cover the 30-year period 1982-2012.² However, this sample has been rather unique, with little major episodes of bond market crises, economic recessions, and inflationary episodes. Since 1980 yield levels displayed a secular decline in most markets. As a result, a key question is how bond factor premiums are influenced by falling or rising yield levels, and other episodes that are typically a concern for investors. In addition, several studies argue that published factor

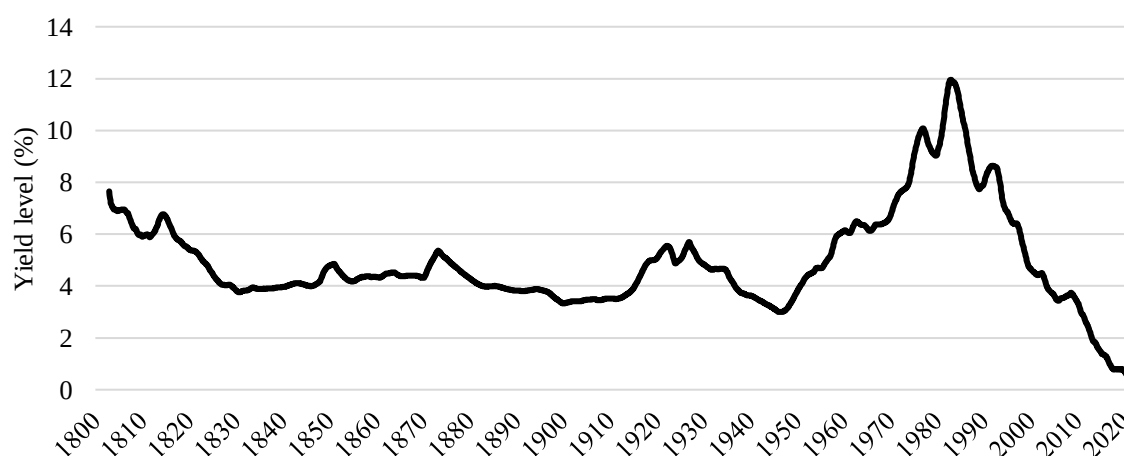
¹ Estimates of the amount of money invested in factor strategies vary from one source to another, ranging from USD 1 to 2 trillion globally in most cases. In a report published in 2017, Morgan Stanley estimates that almost USD 1.5 trillion was invested in smart beta, quant, and factor-based strategies and that assets under management have been growing by 17% per year on average since 2010. According to a survey by FTSE Russell, already 58% of asset owners worldwide have implemented smart beta, - in other words, factor-based strategies - , in their portfolios in 2019.

² In addition, Brooks, Palhares and Richardson (2018) and Kothe, Lohre and Rother (2020) cover Carry, Momentum and Value for Government bonds, but for the even smaller samples 1997 – 2017 and 1994 – 2019, respectively.

premiums could be influenced by ‘p-hacking’ (see Harvey, 2017).³ As a result, published findings might reflect a type I error in testing (i.e., falsely discovering predictability) and may fail to hold out-of-sample.

To address these concerns, we utilize an extensive historical sample that spans all major government bond markets from developed countries over a 221-year period (January 1800 – December 2020). Basically, we have 190 years of additional data to put the published results to the test. In total we have 35,784 monthly return observations in our sample, thereby providing us with sizable testing power to examine bond factor premiums. Moreover, over our sample period global bond yields displayed several secular rates cycles, as illustrated by the development of the global average 10-year yield based on France, Germany, Japan, UK, and US in Figure 1. Roughly post-1980 global government bond yields displayed a secular decline across the world. However, before, yields displayed different behavior, with also secular rises in yields. Thereby, our study also provides a natural robustness test on the influence of secular yield trends on bond factor premiums.

Figure 1: Average 10-year yields 1800-2020



Note: The figure shows the 3-year rolling average of the 10-year yields for France, Germany, Japan, UK, and US from January 1800 to December 2020.

³ *P-hacking* refers to the conscious or unconscious misuse of data analysis to find patterns in data. As a case in point, Harvey, Liu and Zhu (2016) find a clear publication bias pattern in the top finance journals, and that of over 300 documented stock-level anomalies many become questionable after analyzing these in a rigorous testing framework that allows for multiple hypotheses testing bias. P-hacking is not limited to financial economics. P-hacking is mostly discussed in social sciences and medicine. The Economist discussed the topic in 2013 with the headline title: ‘How science goes wrong’. Begley and Ellis (2012) show that out of 53 studies on pre-clinical cancer only 11% could be replicated. An open science collaboration (2015) shows that out of 97 significant psychological studies only 36 could be replicated. In behavioral economics, Camerer et al. (2016) find that out of 18 laboratory studies in economics only 11 can be replicated with similar findings.

In this study we focus on the following three key bond factors: Value, Momentum and Low-Risk, and apply them either across developed market bonds, or within a market on the bond yield curve. These factors are typically considered as key factors within the industry (see for example Blitz, Baltussen and Van Vliet, 2020, and Houweling and Van Zundert, 2017), have been documented in previous studies, and have sufficient coverage over a substantial part of our sample period and across the markets we study. Value and Momentum are applied across bond markets, following Asness, Moskowitz, and Pedersen (2013)⁴, and Low-Risk⁵ is applied on the bond curve, as Frazzini and Pedersen (2014) show Low-Risk is present on the bond curve, but is weak to absent in an across bond market setting. We keep these factors and their definitions unchanged over our out-of-sample period to have a reliable and robust assessment of bond factor premiums, even in the wake of p-hacking.

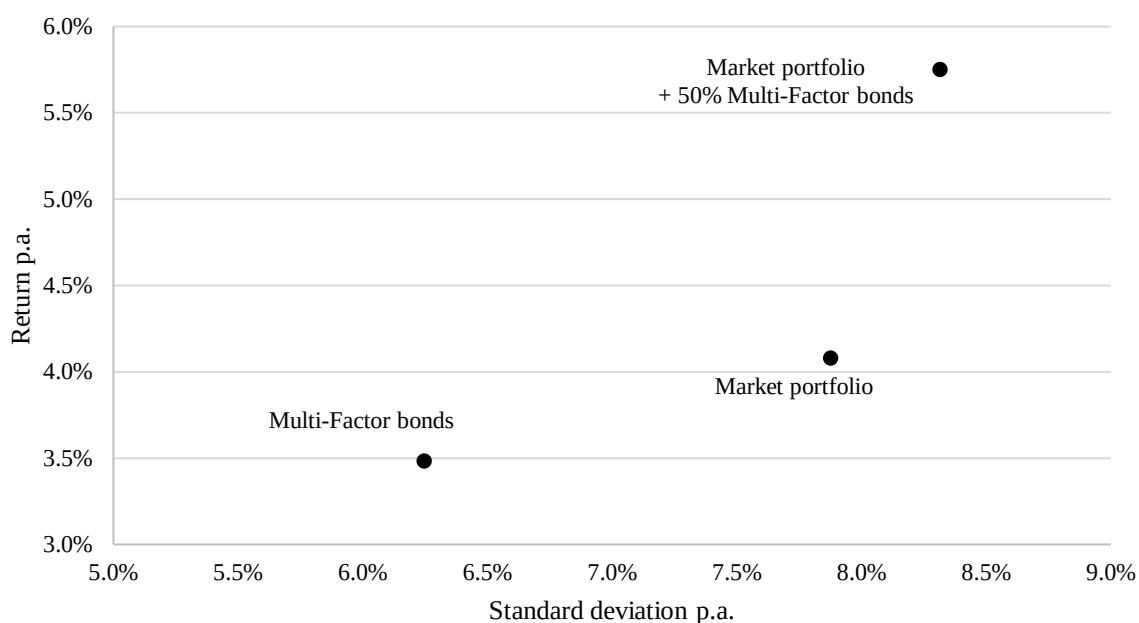
Our findings are as follows. We find that Value, Momentum, and Low-Risk offer attractive factor premiums, with Sharpe ratios of 0.51, 0.24, and 0.40 over our full sample. Moreover, these factor premiums are consistent over time, being positive in 72% (Momentum) to 92% (Value) of 10-year rolling periods. Combining the factors into a simple multi-factor portfolio gives a highly significant Sharpe ratio of 0.56 (t-statistic of 8.22) from 1800 to 2020 and is positive in 89% of the 10-year rolling periods. In other words, factor strategies in government bonds offer attractive returns and diversify each other.

Next, we show a multi-factor bonds portfolio gives robust performance over various macroeconomic states that are typically a concern for investors. These include recessions and expansions, crisis and non-crisis periods, years with either rising or declining yields, years with above or below median inflation, and years with positive or negative equity returns. Note that the previously published papers cover a period with declining yields. It is therefore important to examine performance in years with rising yields and declining yields.

⁴ Asness, Moskowitz, and Pedersen (2013) relate Value to the yield level to inflation ('Real Yield'), and also use the term spread on a bond curve, which is very similar to the definition of Carry covered in Kojien, Moskowitz, Pedersen, and Vrugt (2018). When we refer to Value, we mean both the Real Yield and Carry.

⁵ Frazzini and Pedersen (2014) refer to this phenomenon as Betting-Against-Beta.

Figure 2: Return and risk for the bond market portfolio and bond factor premiums



Note: The figure shows the average annualized returns and standard deviations for the global bond market portfolio ('Market portfolio'), the global Multi-Factor bonds portfolio ('Multi-Factor bonds') – which combines the Momentum, Value and Low Risk factors – and a combination of the bond market portfolio and 50% Multi-Factor bonds. The sample period is 1800-2020.

Finally, we evaluate bond factor premiums in a portfolio context. When considering a long-only bond or multi-asset investor that considers adding bond factor premiums, we find strong value add of a multi-factor bonds portfolio. Figure 2 summarizes the benefits. A multi-factor bonds portfolio has an average return of 3.48% a year at a correlation of -0.05 with the bond market and adding a multi-factor bonds strategy to a passive global government bond portfolio increases returns substantially with little impact on risk. The multi-factor bonds portfolio focuses on country and curve allocation, and hence does not take a stance on the direction of the bond market, but only the relative attractiveness of the different bond markets or maturities. Of course, there are also studies showing that bond market timing can be successful as well, see for example Baltussen, Martens, and Penninga (2021) which provides an out-of-sample test for timing variables introduced by Ilmanen (1995) and others. We find that one of those timing variables which we can test from 1800 onwards, time-series momentum, can further add value to the combination of long bonds and multi-factor bonds.

Our work is closely related to Baltussen, Swinkels, and Van Vliet (2021) who examine global factor premiums across all major asset classes. Compared to their study, we focus on the most

important factors documented in government bond markets: Value and Momentum across bond markets (i.e., ‘country allocation’) and Low-Risk on the bond curve (i.e., ‘maturity selection’). Further, we examine in depth government bond factor premiums, allowing us to share much more insights specific to the bond market, and take an investor perspective, considering both single *and* multi-factor portfolios. In addition, we show how an investor can benefit when moving from passive investing in the bond market to factor-based investing in bonds, both from a long-short and from a long-only investor constraint perspective.

2. Bond factors and data

2.1 Factor definitions

In this study we focus on three key government bond factors, Value, Momentum, and Low-Risk. Asness, Moskowitz, and Pedersen (2013) cover cross-sectional momentum (‘Momentum’) and value (‘Value’) for multiple asset classes. For Momentum we use their definition, which is to compute the past 12-month returns minus the last month. For Value the paper covers three definitions: 5-year mean-reversion, term spread, and the real yield, which is defined as the 10-year bond yield over the past 1-year inflation. Their results for combining the three Value measures are much more promising than the stand-alone results for 5-year mean-reversion. We therefore focus on the real yield (‘Value 1’) and term spread (‘Value 2’) as our Value measures. The term spread, the bond yield minus the risk-free rate, is closely related to carry. The term spread omits the roll-down from Kojien, Moskowitz, Pedersen, and Vrugt (2018) which is not calculable for the long historical dataset we have. For selecting countries, however, they show that the term spread and carry are 94% correlated. The method to construct long-short factor portfolios for both Value measures and Momentum is described in Appendix B.

Finally, we follow Frazzini and Pedersen (2014) for Low-Risk on the U.S. curve. We test this factor by estimating the betas over a 36-month period (requiring at least 12-months of data) for the different maturities. We then go long the low beta maturities, and short the high beta maturities.⁶ The position sizes of each short and long leg are chosen such that the ex-ante betas of both legs are equal, such that the factor has as little overall market effect as possible.

⁶ For a curve strategy this is equivalent to going long the lowest maturity bonds, and short the highest maturity bonds. Note that there is not a long history on durations, and as such it is more convenient to use betas.

2.2 Data

Our dataset is described in detail in Appendix A, including the sixteen developed government bond markets we cover. It also describes the data sources, with Global Financial Data and Macrohistory.net providing the deep history, and Datastream and Bloomberg more recent data. In terms of data quality checks and cleaning data we follow Baltussen, Swinkels and Van Vliet (2021), who extensively analyze data quality and apply multiple data filters to ensure a dataset of good quality. Most importantly, they have applied a number of conservative screens on the data series and removed data points when they do not pass these screens. These screens are (1) a zero-return screen, which leaves out data series with more than one zero or missing spot return observations in the past 12 months, (2) a return interpolation screen, which leaves out identical returns from one month to the next month, and (3) a stale return screen, which leaves out observations that do not have nine or more differentiating returns over the past 12 months. Note that some months, mostly in the 19th century, are missing due to these data filters. Some specific examples for bonds are provided in Appendix A. Momentum and the term spread start in 1800, the real yield, making use of inflation data, starts in 1872. For the Low-Risk strategy on the U.S. curve Global Financial Data provides bond returns for the maturities 1, 2, 3, 5, 10 and 30 years, along with a 3-months short rate series. The shortest three maturities start in 1941, and the longer maturities start in 1919 allowing us to start the Low Risk factor returns from 1922 onwards (as we require a 36-months beta estimation window).

3. The deepest sample evidence on bond factor premiums: 1800-2020

3.1 Recent sample results: Existing evidence and replication

We start our analyses by studying recent sample evidence for the individual government bond factors: Value, Momentum, and Low-Risk. In the left-hand side of Table 1 we show the results the studies mentioned in Section 2.1 document over their sample periods (i.e., Asness, Moskowitz, and Pedersen, 2013, Frazzini and Pedersen, 2014, Koijen, Moskowitz, Pedersen, and Vrugt, 2018). We show the start and end date of their sample period, the variable definition they use, and the Sharpe ratio they report. To that we add a t-statistic based on multiplying the Sharpe ratio by the square root of the number of years in the sample (see Sharpe, 1994). In the right-hand side of Table 1 we show the definitions we use, and the Sharpe ratios and t-statistics we find based on our dataset.

The mentioned studies document Sharpe ratios of 0.06, 0.18, 0.52 and 0.81 for Momentum, Value 1, Value 2, and BAB. The Value 2 and BAB Sharpe ratios are statistically significant

with t-statistics of 2.79 and 6.26. It is good to put the size of the Sharpe ratios into perspective. Dimson, Marsh and Staunton (2020) report Sharpe ratios of 0.25 and 0.11 over the period 1900-2020 for world equity and bond markets, respectively. Note that we would need 61 years to get a t-statistic of 1.96 for a Sharpe ratio of 0.25.

Table 1: Replication of existing studies for government bond factor premiums

	From existing studies					Our replication		
	Start date	End date	Definition	Sharpe	t-stat	Definition	Sharpe	t-stat
Momentum	198201	201107	12M-1M	<i>0.06</i>	0.35	12M-1M	0.15	0.82
Value 1	198201	201107	5Y Reversal	<i>0.18</i>	0.97	Real Yield	0.29	1.59
Value 2	198311	201209	Slope+Roll	<i>0.52</i>	2.79	Term Spread	0.47	2.52
Low-Risk	195302	201203	Beta	<i>0.81</i>	6.26	Beta	0.41	3.12

Note: The table contains sample start and end dates (in format yyyyymm), factor definitions ('Definition'), Sharpe ratios ('Sharpe'), and t-statistics ('t-stat') for each of the original studies on global factor premiums: Asness, Moskowitz, and Pedersen (AMP; 2013) for cross-sectional momentum ('Momentum') and value ('Value 1' & 'Value 2'), and Frazzini and Pedersen (2014) for Low-Risk. AMP use 5Y Reversal for their main results, and show results for the combination of 5Y Reversal, Real Yield, and Term Spread (but not separately). We therefore base Value 2 on Koijen, Moskowitz, Pedersen, and Vrugt (2018). Sharpe ratios in italics are taken from these studies based on their results for government bonds or bond futures. The last three columns are based on our own calculations based on our own data, see Appendix A for details.

We perform a replication exercise on these studies based on our sample. As mentioned before, we use the same definitions for Momentum and Low-Risk, and use real yield and term spread as Value measures. There are also some important differences between the existing studies and our replication exercise. Most notably, we cover 16 countries compared to 10 countries in Asness et al. (2013). The six additional countries are Belgium, France, Italy, Netherlands, Spain, and New Zealand. In the right-hand side of Table 1 we show the results.

We find a higher Sharpe ratio for Momentum (0.15 versus 0.06). For Value 1 (i.e., real yield) we find a Sharpe ratio of 0.29 compared to the reported 0.18 for 5-year mean-reversion, while we find a Sharpe ratio of 0.47 for Value 2 (i.e., term spread), similar to the 0.52 from Koijen et al. (2018). For Low-Risk we get a lower Sharpe ratio of 0.41 compared to 0.81 in Frazzini and Pedersen (2014), albeit both highly significant. Frazzini and Pedersen use the CRSP Fama bond maturity portfolios covering 1 to 5-year maturities, whereas we also include 10- and 30-year maturities. This means that we look at long 1-2-3-year versus short 5-10-30-year bonds,

whilst Frazzini and Pedersen look at 1-2-year versus 4-5-year bonds. Hence large differences are expected. Due to 10- and 30-year bonds we can cover a longer history and covering the full range of maturities also makes sense from a practitioner's perspective as bond indices also cover the full range of maturities.

3.2 Deep-sample sample evidence

Our dataset allows for a very large out-of-sample test for the numbers on the right-hand side in Table 1. We look at both data from 1800 until the start date of the global factor premium studies as provided in column 3 of Table 1; and at newer data from the end dates in Table 1 until the end of 2020. The results are provided in Table 2.

Table 2: Bond factor premiums: 1800-2020

Factor	Measure	Replication		Deep sample		Full sample	
		Sharpe	t-stat	Sharpe	t-stat	Sharpe	t-stat
Momentum	12M-1	0.15	0.82	0.25	3.38	0.24	3.46
Value 1	real yield	0.29	1.59	0.18	1.98	0.20	2.44
Value 2	term spread	0.47	2.52	0.56	7.64	0.54	8.02
Low-Risk	Beta	0.41	3.12	0.38	2.29	0.40	3.85

Note: For 12M-1M momentum ('Momentum'), the real yield ('Value 1'), the term spread ('Value 2') and beta ('Low-Risk') we look at Sharpe ratios ('Sharpe') and t-statistics ('t-stat') in three samples. The 'Replication' sample period covers the replication results over the sample period covered by the global factor premiums studies Asness et al. (2013), Koijen et al. (2018), and Frazzini and Pedersen (2014), see Table 1. The 'Deep sample' period covers earlier data starting in January 1800 (Momentum and Value 2), 1872 (Value 1) or 1922 (Low-Risk), and newer data ending in December 2020, but leaving out the replication-sample data. The 'Full sample' period covers all data.

The deep-sample test generally shows strong results. For Momentum we see a highly significant Sharpe ratio of 0.25, higher than the 0.15 for the 1982-2011/07 replication-sample period. For Value based on real yields the Sharpe ratio is 0.18, lower than the 0.29 for the replication-sample period. It is, however, still significant at the five percent significance level. For Value based on the term spread the Sharpe ratio is a highly significant 0.56 for the deep-sample, compared to 0.47 for 1983/11-2012/09. Finally, the Sharpe ratio for Low-Risk is 0.38 (t-statistic 2.29), quite similar to the 0.41 replication -sample Sharpe ratio. The final two columns in Table 2 show the full-sample results. For all four factors we see significant Sharpe ratios ranging from 0.20 for Value 1 to 0.54 for Value 2, with generally high t-statistics (from 2.44 to 8.02).

3.3 Multi-factor bonds

Next, we examine the combination of the above individual factors into a multi-factor portfolio. To this end, we combine the four bond factor returns using an equal weighting scheme:

$$R_{Multi-Factor,t} = 0.25 R_{Momentum,t} + 0.25 R_{Value\ 1,t} + 0.25 R_{Value\ 2,t} + 0.25 R_{Low-Risk,t} \quad (1)$$

Before the start of Low-Risk in 1922 we use an equal weighted combination of the other factors. A remark is in order about the weight of Value, as one could argue that Value gets a higher weight than Momentum and Low Risk. Our motivation is simplicity combined with having two Value measures which are quite distinct, as their correlation equals -0.18. Overall, correlations between factor returns are limited. The highest correlation is 0.34 between Momentum and Value 2, which is related to the fact that Momentum is based on total returns, and these returns depend on both the term spread (our Value 2 measure) and yield changes. The lowest correlation is -0.39 between Momentum and Value 1. The correlations between Low Risk and the other three components in equation (1) are very close to zero. The results that follow are robust to using different weighting schemes to the individual factors, being qualitatively similar when we first build one value basket and then give 1/3rd weight to each factor, or when using mean-variance optimal weights. Table 3 shows the results of the multi-factor bonds combination over our full sample period (1800-2020) and over the subsample when Low-Risk is available (1922-2020).

Multi-Factor bonds has a highly significant Sharpe ratio of 0.56 for 1800-2020, and 0.72 for 1922-2020 when also Low-Risk is available. Due to the aforementioned low correlations between the four individual factors, we see that Multi-Factor has a substantially lower annualized standard deviation of 6.25%, compared to the average of 11.22% for the individual factors. As mentioned in Appendix B we always ensure an ex-ante volatility of 10% for each individual factor at the start of each month. Ex-post we see that volatilities for the individual factors are just above 10%.

Table 3: Multi-Factor bonds

	Momentum	Value 1	Value 2	Low-Risk	Multi-Factor
<i>Panel A: 1800-2020</i>					
Return p.a.	2.58%	2.39%	5.84%	-	3.48%
Stdev p.a.	10.93%	11.94%	10.73%	-	6.25%
Sharpe ratio	0.24	0.20	0.54	-	0.56
t-statistic	3.46	2.44	8.02	-	8.22
<i>Panel B: 1922-2020</i>					
Return p.a.	1.81%	3.38%	6.71%	4.46%	3.98%
Stdev p.a.	11.42%	11.18%	11.42%	11.27%	5.53%
Sharpe ratio	0.16	0.30	0.59	0.40	0.72
t-statistic	1.58	3.01	5.83	3.85	7.16

Note: In this table we examine the results for 12M-1M Momentum ('Momentum'), real yield ('Value 1'), term spread ('Value 2') and Beta ('Low-Risk') for the 1800-2020 and the 1922-2020 sample periods. In addition, we look at the equally weighted average, 'Multi-Factor'. We show average returns, standard deviations, Sharpe ratios and t-statistics per factor and their combination for two sample periods. Momentum and Value 2 start in 1800, Value 1 in 1872 and Low-Risk in 1922.

3.4 Persistence over time

Next, we examine the robustness of bond factor premiums over time. To this end we first examine the performance over rolling 10-year subperiods. Table 4 summarizes the results in terms of success ratio (i.e., number of 10-year period with positive performance). The success ratio of all four factors is at least 72%, i.e., the factors have a positive performance in at least 72% of the rolling 10-year subperiods. Multi-Factor bonds has a success ratio of 89%.

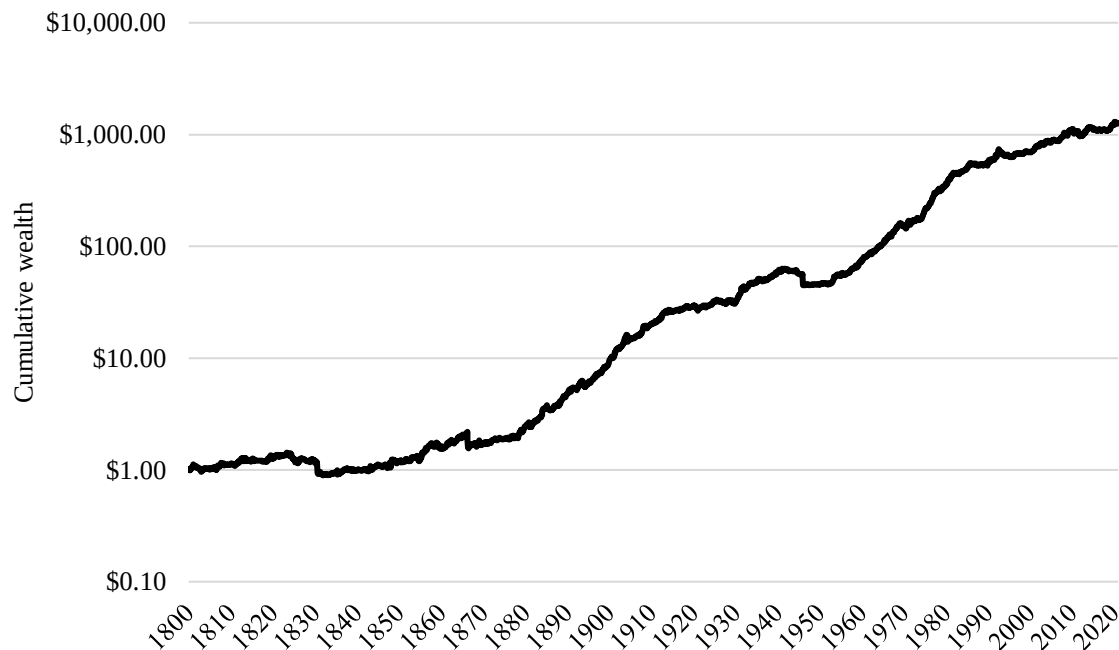
Table 4: Success ratios in 10-year rolling periods

	Momentum	Value 1	Value 2	Low-Risk	Multi-Factor	Market
Success ratio	72.1%	74.3%	87.7%	92.3%	89.0%	91.9%
Nr of obs.	2533	1786	2533	1188	2533	2533

Note: In this table we report the success ratio and number of rolling 10-year subperiods ('Nr of obs.') of the individual bond factors, their equally weighted Multi-Factor combination, and the bond market ('Market'). The success ratio is computed by taking the performance over rolling 10-year (120-month) subperiods and counting how often these are positive, which is subsequently divided by the total number of observations.

Second, we look at the cumulative performance over time in Figure 3. We see that, apart from a modest start in the first decades of the 19th century, when the universe consists of only 7 out of the in total 16 countries, performance is stable over time.

Figure 3: Cumulative performance multi-factor bonds



Note: The figure shows how the cumulative wealth on an initial investment of 1 USD at the end of 1799 of investing in the Multi-Factor bonds portfolio that combines Momentum, Value and Low Risk government bond factors.

4. Market risk and factor returns across ‘good’ and ‘bad’ states

In this section we look in more detail at the return and risks of factor premiums relative to the bond market and across market and macroeconomic states. To this end we utilize our full sample of 221 years of data, as this gives us a substantial number of observations across good and bad states compared to a sample of for example the most recent 30 years. We first regress the returns of multi-factor bonds on the global government bond market portfolio, which we proxy by the equal-weighted return over the bond markets included in our sample. Even though we are looking at cross-sectional strategies, they may still benefit from a bottom-up structural bond beta. More specifically, we run the following regression:

$$R_{Multi-Factor,t} = \alpha + \beta \cdot R_{Global\ Market,t} + \varepsilon_t \quad (2)$$

Table 5 summarizes the results by means of the beta, the Appraisal Ratio (i.e., the estimated alpha divided by the standard deviation of the residuals of equation (2)), and its t-statistic. We present the multi-factor bonds results and the results for the individual factors, by replacing the

Multi-Factor returns in equation (2) by the individual factor returns. In general, we see betas close to zero, indicating there is no structural positive beta that could explain the performance of government bond factor strategies. Because the betas are small, the Appraisal Ratios are close to the Sharpe Ratios in Table 3. A negative beta leads to an Appraisal Ratio which is slightly higher than the Sharpe ratio, and vice versa. Because Multi-Factor has a small negative beta, the Appraisal ratios at 0.59 for 1800-2020 and 0.75 for 1922-2020 are slightly higher than the Sharpe ratios of respectively 0.56 and 0.72.

Table 5: Appraisal ratios

	Momentum	Value 1	Value 2	Low-Risk	Multi-Factor
<i>Panel A: 1800-2020</i>					
Beta	-0.02	0.06	-0.07	-	-0.04
Appraisal ratio	0.24	0.18	0.57	-	0.59
t-statistic	3.58	2.25	8.39	-	8.63
<i>Panel B: 1922-2020</i>					
Beta	0.01	0.04	-0.07	-0.15	-0.03
Appraisal ratio	0.15	0.29	0.61	0.45	0.75
t-statistic	1.53	2.86	6.07	4.41	7.41

Note: We regress the factor returns on the Global Market, the average of the individual bond market returns. The table shows the beta of the regression, the Appraisal Ratio (i.e., the alpha of the regression divided by the standard deviation of the residuals), and its t-statistic for the 1800-2020 and the 1922-2020 sample periods.

Next, we examine whether perhaps the Multi-Factor bond returns are poor in ‘bad’ states as a possible explanation for the good performance in the long run. For this we assign each of the calendar years to a ‘bad’ or a ‘good’ state based on:

- Recessions and Expansions
- Crisis and Non-Crisis
- Negative and positive equity performance
- Yields rising or declining
- Inflation above or below the median

Appendix A provides detail on the source of the classifications. Table 6 shows the results.

Multi-Factor bonds provides on average positive returns in both bad and good states. In Recessions the average return is 1.9%, in Crisis periods 4.1%, when yields rise 3.9%, when inflation is high 4.1%, and when equity markets have negative returns 3.1%. Hence based on

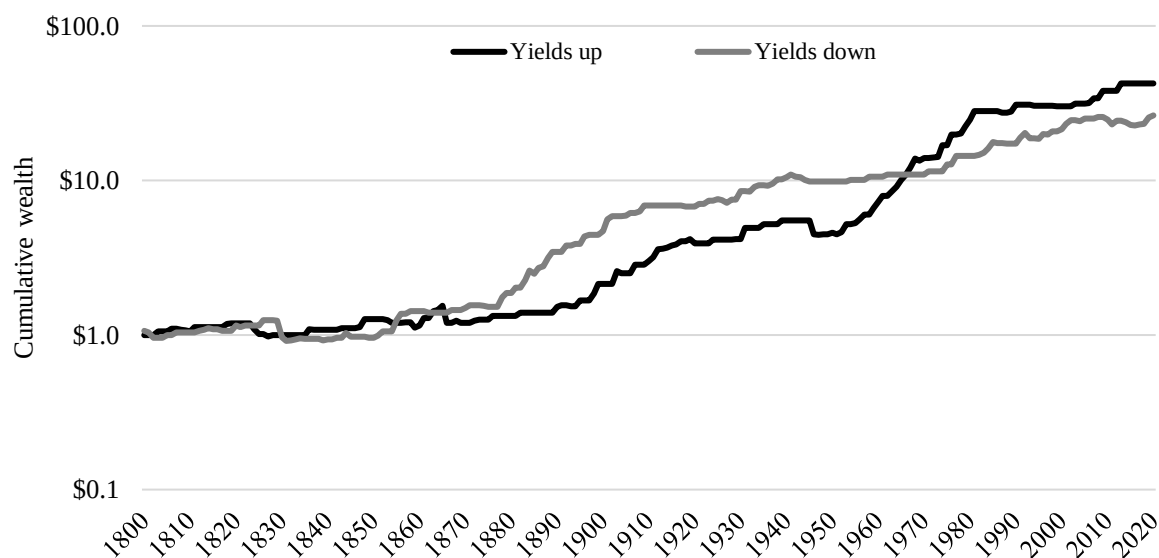
this evidence the returns of Multi-Factor bonds cannot be explained by poor performance in ‘bad states’. From this we can conclude that a risk-based explanation for the factor premiums seems unlikely.

Table 6: Performance Multi-Factor bonds in ‘good’ and ‘bad’ states

All	Growth		Crisis		Yields		Inflation		Equity	
	recession	expansion	yes	no	up	down	high	low	negative	positive
221	55	166	75	146	103	118	74	78	44	177
3.5%	1.9%	3.9%	4.1%	3.1%	3.9%	3.0%	4.1%	4.9%	3.1%	3.5%

Note: We divide all calendar years into two groups per indicator: recession or expansion; crisis or no crisis; yield up or down; and negative or positive equity market performance. The first row shows the number of calendar years in each category. The second row shows the performance of Multi-Factor bonds. The sample period is 1800-2020. Inflation data only starts in 1869 and hence covers 152 years instead of 221 years.

Figure 4: Performance Multi-Factor bonds in rising and declining yields



Note: The figure shows how the cumulative wealth on an initial investment of 1 USD at the end of 1799 of investing in the Multi-Factor bonds portfolio that combines Momentum, Value and Low Risk government bond factors conditional on yields up or yields down states. Each of the 221 calendar years is labelled either ‘yields up’ when the average yield of all the markets increased; or ‘yields down’ when the average yield declined.

Perhaps most importantly for investors: The performance of multi-factor bonds is stable across the different scenarios, including for example falling or rising equity markets (thereby providing evidence of added value over equity markets), or high and low inflation episodes.

Interestingly, performance is good regardless of yields rising or falling. Our deep historical sample makes it possible to make such a statement, as it also contains multi-year periods with also rising yields, unlike a sample that covers post-1980 results. As mentioned before, the key academic studies cover a period from 1982 onwards with mainly declining yields. To provide more color on performance during rising and declining yield period we show the cumulative performance of Multi-Factor bonds conditional on both scenarios in Figure 4. Performance is consistent over time across both scenario's. Moreover, results not shown in the table reveal that all factors contribute positively in both rising and declining yield calendar years. Especially the second value factor (term spread) and low risk perform strongly in periods with rising yields.

5. The added value of bond factor premiums to a bond market portfolio

So far, we have presented the results of long-short (zero-investment) portfolios, as is common in academic studies. Next, we examine what happens when we add the long-short multi-factor bonds portfolio to a long-only passive bond market portfolio that invests equally weighted in all government bond markets. Table 7 shows the results. The return-risk ratio of the bond portfolio can be improved from 0.52 to 0.69 (adding 50% Multi-Factor bonds) or 0.77 (adding 100% Multi-Factor bonds).

Table 7: Combining a passive bond market portfolio with factor premiums

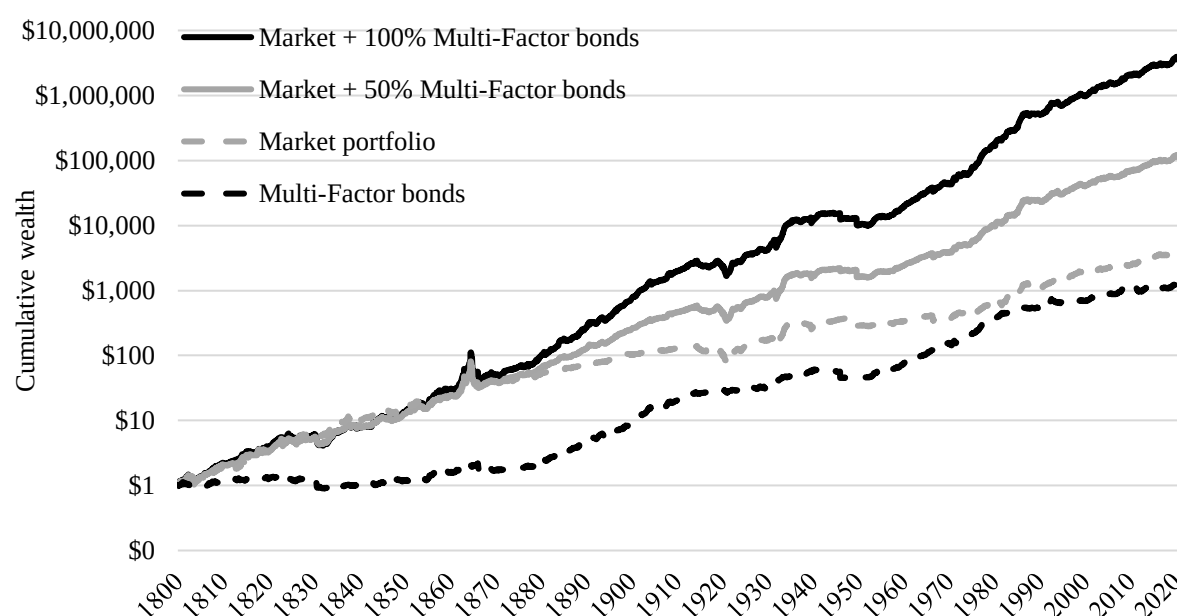
	Market portfolio	Multi-Factor bonds	Market + 50% Multi-Factor bonds	Market + 100% Multi-Factor bonds
Return p.a.	4.08%	3.48%	5.75%	7.49%
Stdev p.a.	7.88%	6.25%	8.32%	9.79%
Return-risk ratio	0.52	0.56	0.69	0.77
Yields up	0.42%	3.94%	2.34%	4.31%
Yields down	7.27%	2.98%	8.55%	10.04%

Note: We combine the buy-and-hold bond market portfolio, that invests equally weighted in all bond markets ('Market portfolio'), with the (long-short) Multi-Factor bonds ('Multi-Factor bonds'). Shown are the average returns, standard deviations, and return/risk ratios of the market portfolio, Multi-Factor bonds, the bond market portfolio plus 50% multi-factor bonds, and the bond market portfolio plus 100% multi-factor bonds. In the last two rows we divide the calendar years into years where yields went up and years where yields declined, as defined in Table 6, and show average returns per scenario. The sample period is 1800-2020.

The bottom two rows of Table 7 splits these results in the years of rising or declining yields. The bond market portfolio on average returns 0.42% in years where yields increase, compared

to 7.27% when yields decline. Adding Multi-Factor makes calendar years with rising yields substantially more attractive for investors. The same holds for years with declining yields. Hence a bond market investor would clearly benefit from using Multi-Factor bonds. Figure 5 shows the cumulative performance of the four cases in Table 7. Finally, we like to note that we have verified that the added value of factor premiums to a passive bond portfolios is robust to different portfolio construction approaches for multi-factor bonds, such as tercile portfolios, two portfolios, or lower weights (up to 10%) to multi-factor bonds, approaches that actively prevent short positions and enforce a long-only portfolio.

Figure 5: Cumulative performance long bonds with multi-factor overlay



Note: The figure shows how the cumulative wealth on an initial investment of 1 USD at the end of 1799 of investing in various bond portfolios. We combine the buy-and-hold bond market portfolio, that invests equally weighted in all government bond markets ('Market portfolio'), with the (long-short) multi-factor bonds ('Multi-Factor bonds'). Shown are the cumulative performance of the bond market portfolio, multi-factor bonds, the bond market portfolio plus 50% multi-factor bonds, and the bond market portfolio plus 100% multi-factor bonds.

So far, we analyzed bond factors that do country allocation or curve allocation, so-called cross-sectional strategies. Such strategies by nature do not take a stance on the direction of the bond market; they only assess the relative attractiveness of the different bond markets or maturities. Of course, there are also studies showing that bond market timing can be successful as well, see for example Baltussen, Martens, and Penninga (2021) which provides an out-of-sample test for timing variables introduced by Ilmanen (1995) and others. Therefore, we next examine whether cross-sectional Multi-Factor bonds can add value to bond market timing. To this end,

we take as an example one timing variable that is available from 1800, time-series momentum. Both cross-sectional and time-series momentum are based on 12-months momentum, skipping the most recent month. But instead of building a long-short portfolio time-series momentum takes a long or short position in each market based on the sign of the return, following Moskowitz, Ooi and Pedersen (2012). Hence potentially it is long in all bond markets, or short in all bond markets.

Table 8 shows the results for the bond market portfolio, Multi-Factor bonds, and Time-Series Momentum, as well as combining these strategies by adding 50% Multi-Factor bonds and 25% Time-Series Momentum to the passive bond market portfolio. We apply a lower weight for Time-Series Momentum to account for its about factor 2 higher volatility compared to the diversified Multi-Factor bonds portfolio (we have verified that more sophisticated techniques like mean-variance optimizations lead to qualitatively similar conclusions). The correlation between Multi-Factor and Time-Series Momentum is relatively low (0.19). Hence it is no surprise the risk-return ratio goes up from 0.69 – when combining the bond market portfolio with 50% Multi-Factor bonds – to 0.82 when also adding 25% Time-Series Momentum in government bonds. Hence both Multi-Factor bonds and time-series momentum add value on top of each other and relative to a passive bond market portfolio.

Table 8: Combining a passive bond market portfolio with factor premiums and time-series momentum

	Market portfolio	Multi-Factor bonds	TS Momentum	Combination
Return p.a.	4.08%	3.48%	6.48%	7.39%
Stdev p.a.	7.88%	6.25%	11.68%	9.00%
Return-risk ratio	0.52	0.56	0.55	0.82

Note: We combine (‘Combination’) the buy-and-hold bond market portfolio, that invests equally weighted in all bond markets (‘Market portfolio’), with 50% of Multi-Factor bonds (‘Multi-Factor bonds’) and 25% Time-Series Momentum (‘TS Momentum’). Shown are the average returns, standard deviations, and return/risk ratios of the market portfolio, Multi-Factor bonds, Time-Series Momentum, and their combination. The sample period is 1800-2020.

6. Concluding remarks

We extensively study factor premiums in global government bond markets over a deep sample of 221 years of data between 1800 and 2020. Existing bond factor studies only cover the 1982-2012 period, which is characterized by strongly declining yields, and is subject to potential p-hacking concerns. Our findings reveal that bond factors (Value, Momentum, Low-risk) offer attractive premiums that do not decay across samples, are persistent over time, and consistent across various market or macroeconomic scenarios. As such we provide both deep-sample evidence and the important insight that bond factor performance is also strong in periods of rising yields. A multi-factor bonds strategy that combines Value, Momentum and Low Risk provides the strongest risk-adjusted returns and has a stable performance over time that is strong regardless of being in 'good' or 'bad' states as characterized by expansions & recessions, non-crises or crises, positive or negative equity returns, or low or high inflation. Further, the factor premiums diversify to each other, as well as to bond or equity market risks, and consistently add value over a bond market portfolio. Overall, a multi-factor bonds portfolio is interesting for bond investors, as it robustly adds value over a passive government bond portfolio.

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Appendix A: Historical database construction

We have compiled our data from several sources in order to obtain a reliable and historically extensive dataset. Our sample covers 221 years of data from 31 December 1799 through 31 December 2020.

Bond data

We source bond futures price and return data from Bloomberg, splice these with bond index-level data from Datastream, backfilled before inception with Global Financial Data (GFD). From the same sources we obtain yields, and inflation data, the latter extended where possible with data from Macrohistory.net. We apply a two-months lag to inflation numbers of mimic their real-time availability. The markets we consider are the major developed bond markets around the globe. Table A.1. summarizes the start dates of the data series in our sample.

Table A1: Bond market data

Bond market	GFD	BB futures	DS
Australia	1857-1986	1987-2020	1987-2020
Belgium	1831-2020		
Canada	1953-1984	1989-2020	1985-2020
Denmark	1800-2020		
Germany	1800-1979	1990-2020	1980-2020
France	1800-1984	2012-2020	1985-2020
Italy	1807-1990	2009-2020	1991-2020
Japan	1870-1981	1985-2020	1982-2020
Netherlands	1800-1979		1980-2020
Norway	1822-1991		1991-2020
Spain	1800-1989		1990-2020
Sweden	1853-1986		1987-2020
Switzerland	1900-1979		1980-2020
United Kingdom	1800-1979	1982-2020	1980-2020
United States	1800-1979	1982-2020	1980-2020
New Zealand	1861-1990		1991-2020

Note: GFD = Global Financial Data; BB = Bloomberg; DS = Refinitiv Datastream. The GFD and DS data are 10-year bond returns and bond yields (DS: 7-10y maturity bucket). From BB we get 10-year bond futures returns.

Economics

We construct our global recession data from a splicing of the OECD G7 recession indicator from the OECD website (1960-2020), the NBER US recession indicator from the NBER website (1864-1959), and the contraction of real-GDP from GFD (1800-1863). We obtain the historical data on crisis periods from Carmen Reinhart and Kenneth Rogoff, using their Banks, Currency, Default, Inflation (BCDI) index, which starts in 1800.⁷

Data Quality

The deep historical data tends to be of lesser quality compared to the more recent data, as digital archives, and the use of indices with strong requirements on data processes did not exist. Instead, data was maintained typically by exchanges, statistical agencies, newspapers, and investors annuals, often in manual writing. For specific issues concerning historical data for government bonds, we refer to Baltussen, Swinkels and Van Vliet (2021) for more detail.

To construct a high-quality data set we build upon Baltussen, Swinkels and Van Vliet (2021). They have taken the following three steps. First, they have checked and corrected each data series on potential data errors (see ‘Data cleaning procedure’). Second, they have verified the data sources, when possible, against other sources and found that average returns and volatilities are generally of comparable magnitudes across databases. Third, they have applied a number of conservative screens on the data series and removed data points when they do not pass these screens. These screens are (1) a zero-return screen, which leaves out data series with more than one zero or missing spot return observations in the past 12 months, (2) a return interpolation screen, which leaves out identical returns from one month to the next month, and (3) a stale return screen, which leaves out observation that do not have nine or more differentiating returns over the past 12 months. The first screen filters for data historically available at a non-monthly frequency and for reduced liquidity. The second screen filters an unlikely return pattern, exactly identical consecutive monthly returns, which indicate return interpolation. The third screen filters returns not updated at the monthly frequency. To this end, they remove an asset when, over the past 12 months, fewer than nine unique monthly returns were rounded to 5 basis points. They have simulated that such a pattern is unlikely under a normal or in the replication-sample return distribution for all asset classes in our universe. Further, following Baltussen, Swinkels and Van Vliet (2021) we skip a month between the

⁷ <http://www.reinhartandrogoff.com/data/>

momentum signals and investing, which removes possible spurious autocorrelation at the monthly frequency. We like to stress that these screens mitigate data quality concerns but could bias factor premium estimates downward if they remove correct data points.

Data cleaning procedure

We have taken the following steps to check the quality of each data series and clean for obvious measurement errors. First, we run a studentized residuals outlier test on each series to capture potential data errors, and visually checked each series on jumps and outliers. Potential outliers are manually verified (by comparing to other data sources where possible and by searching for reasons for large price moves), and when due to a data error corrected. The correction in pre sample data points include:

- Dutch government bond returns have a misprint on 1964-11-30, which we replaced by the approximation based on changes in yields times duration.
- Several systematic corrections for financing rates and bond yields, as some yields switch between decimals and percentage notations.

Appendix B: Portfolio construction procedure

After obtaining the factor measures per bond market we construct factor investment portfolios at the end of every month in the following manner. We rank the markets based on the factor measure and take a position equal to the rank minus its cross-sectional average (requiring a minimum of two markets to be present). This procedure is similar to that used by Asness, Moskowitz, and Pedersen (2013), Frazzini and Pedersen (2014), and Kojen, Moskowitz, Pedersen, and Vrugt (2018). Consequently, positions for all factors add up to zero at each point in time:

$$w_t^i = z_t \cdot \left(\text{Rank}(S_t^i) - \frac{N_t + 1}{2} \right),$$

with w_t^i the weight of asset i at time t , S_t^i the factor signal, N_t the number of assets in the cross-section, and z_t a scaling factor to ensure that the portfolio sums to zero.

Next, we size positions in each market by its simple 3-year rolling volatility estimate or beta estimate (for BAB only), in the same spirit as Asness, Moskowitz, and Pedersen (2013) and Frazzini and Pedersen (2014), but fitted to our sample frequency (i.e., monthly data). To prevent undue impact from extremely low volatility estimates (and hence keep the factor strategy robust from an investor perspective), especially in the earlier part of our sample, we floor each volatility (beta) estimate at the maximum of the 10% quantile of volatility (beta) estimates or 2.5% (0.25), whichever is greater.

We subsequently sum the product of position, sizes, and market returns across markets for each date to generate the return on the factor strategy. We then adjust the position sizes of each of the factor strategies using a 10-year rolling window such that each factor strategy targets an ex-ante volatility of 10% per annum. This approach takes an ex-ante view of portfolio construction, as available in real time. However, our results are not materially different if we simply scale by the sample ex-post volatility, as in Kojen, Moskowitz, Pedersen and Vrugt (2018).

We rebalance the portfolios each month based on the signals and volatility estimates. This methodology results in balanced long-short portfolios.